

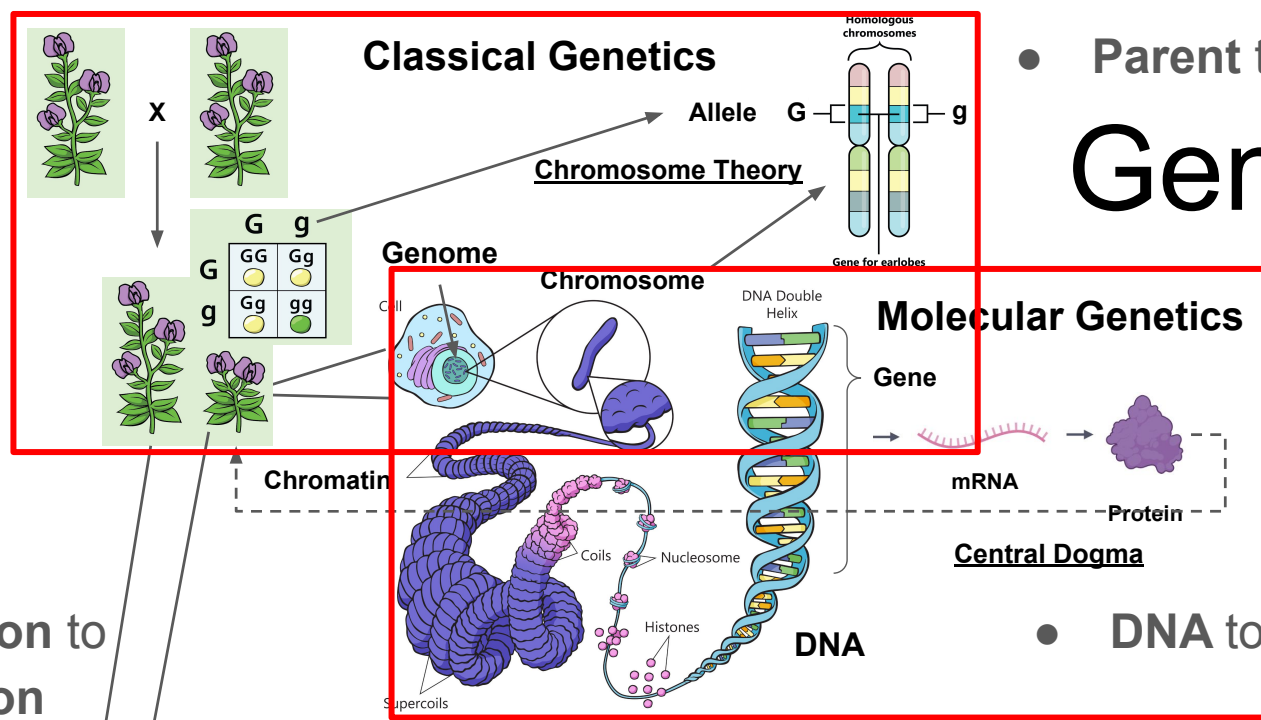
NCBI Database and BLAST

SCiUS

Lect. Todsapol Techo

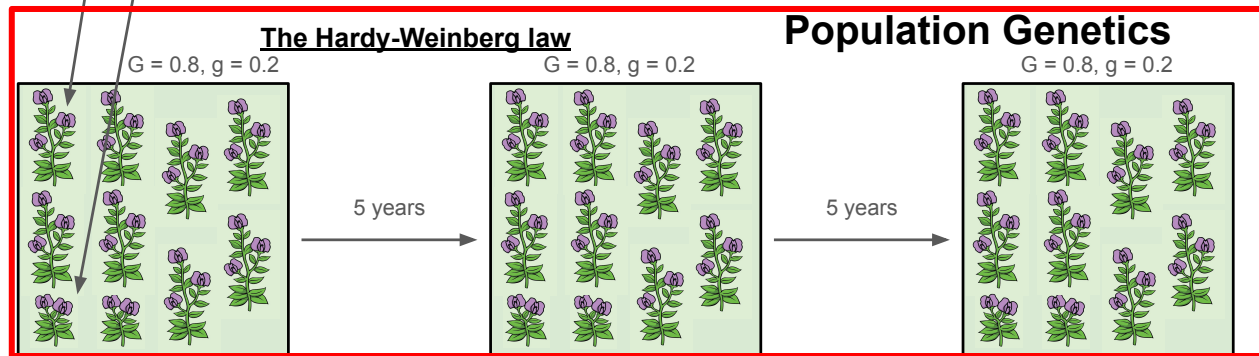
Genetics

- Parent to offspring



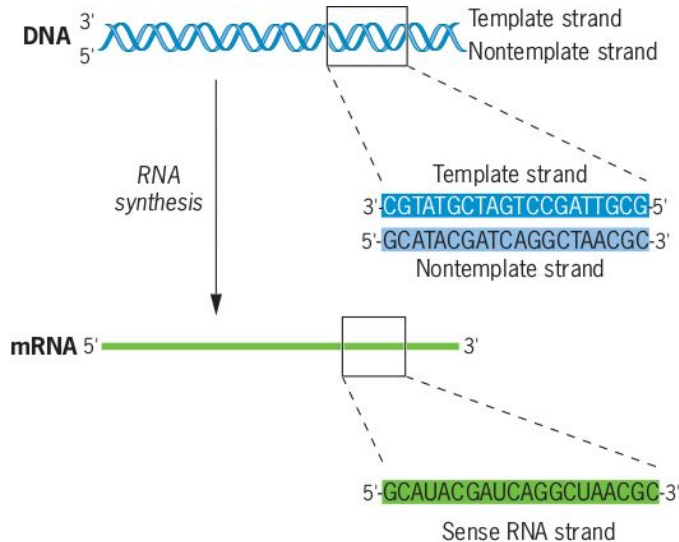
- Generation to generation

- DNA to gene action

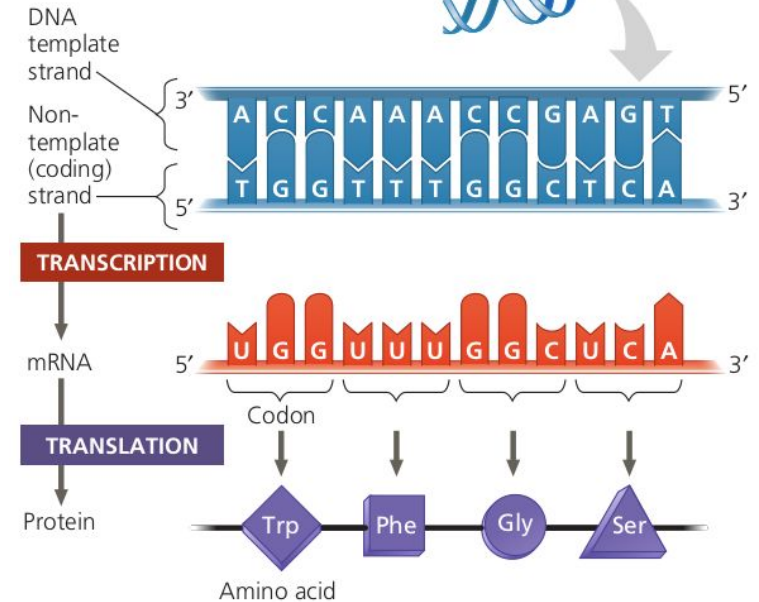


Genetic Information and Genetic Code

- Template Strand (Antisense DNA)
- Nontemplate or RNA-like Strand or (sense DNA)
- Triplet Nucleotides (Triplet Code)

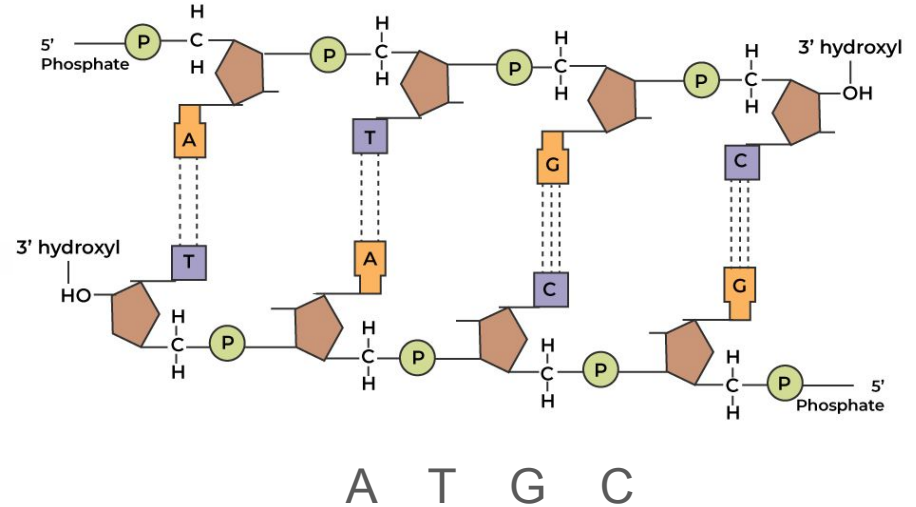
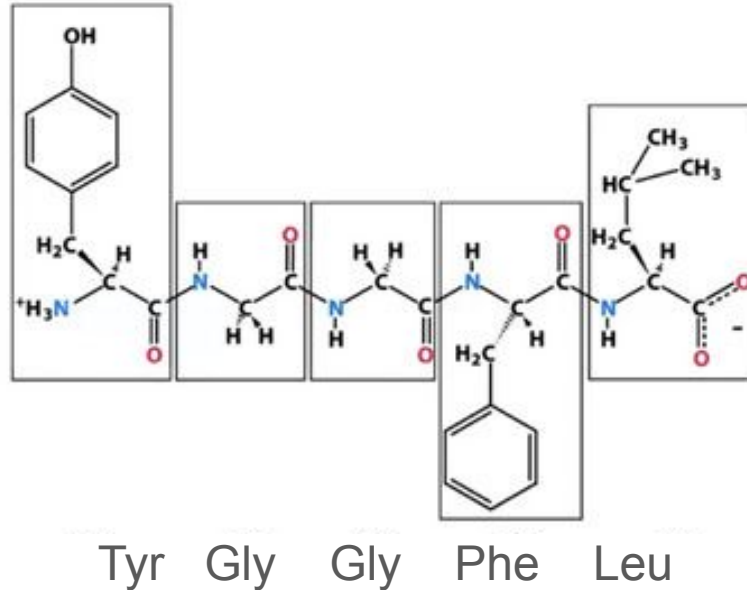


▼ **Figure 17.5 The triplet code.** For each gene, one DNA strand functions as a template for transcription of RNAs, such as mRNA. The base-pairing rules for DNA synthesis also guide transcription, except that RNA is made with uracil (U) instead of thymine (T). During translation, the mRNA is read as a sequence of nucleotide triplets, called codons. Each codon specifies an amino acid to be added to the growing polypeptide chain. The mRNA is read in the 5' → 3' direction.



Biological Sequence

- Polynucleotide: ATGTGCTAATCCTTGGT
- Polypeptide: Met-Lys-Thr-Ile



How to obtain Biological Sequences from living?

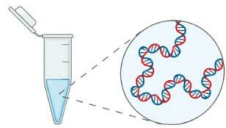
- **DNA sequencing**

- Sanger Sequencing (Low throughput)
- Next Generation Sequencing (NGS) (High throughput)

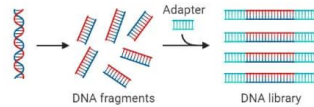
- **Protein sequencing**

- Edman Degradation (Low throughput)
- Tandem Mass Spectrometry (MS/MS) (High throughput)

Step 1:
DNA extraction

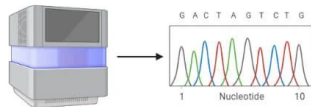


Step 2:
Library preparation

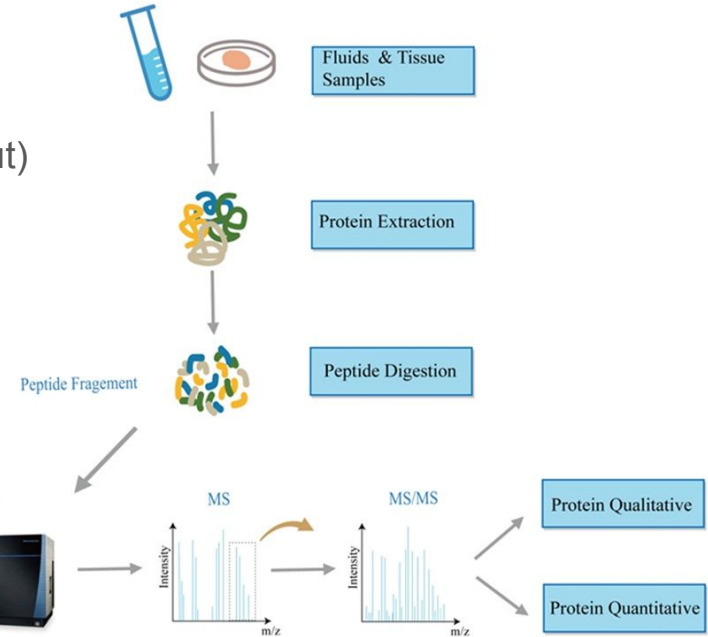


Next Generation Sequencing Workflow

Step 3:
Sequencing

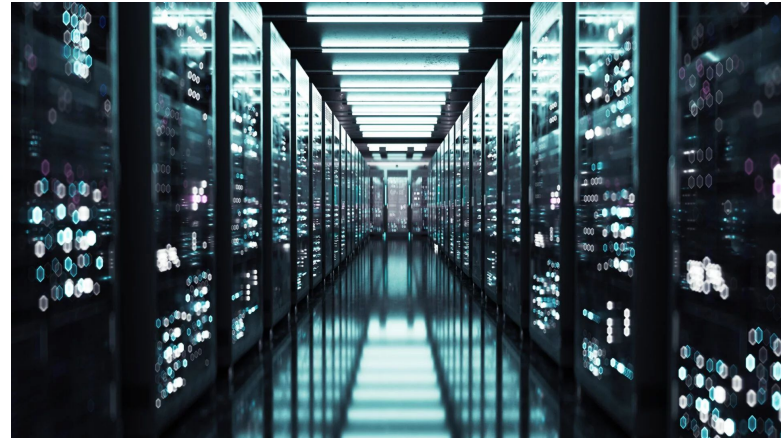
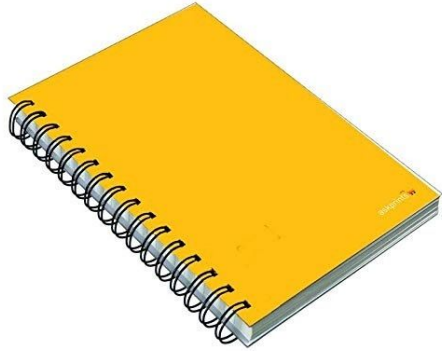


Step 4:
Analysis



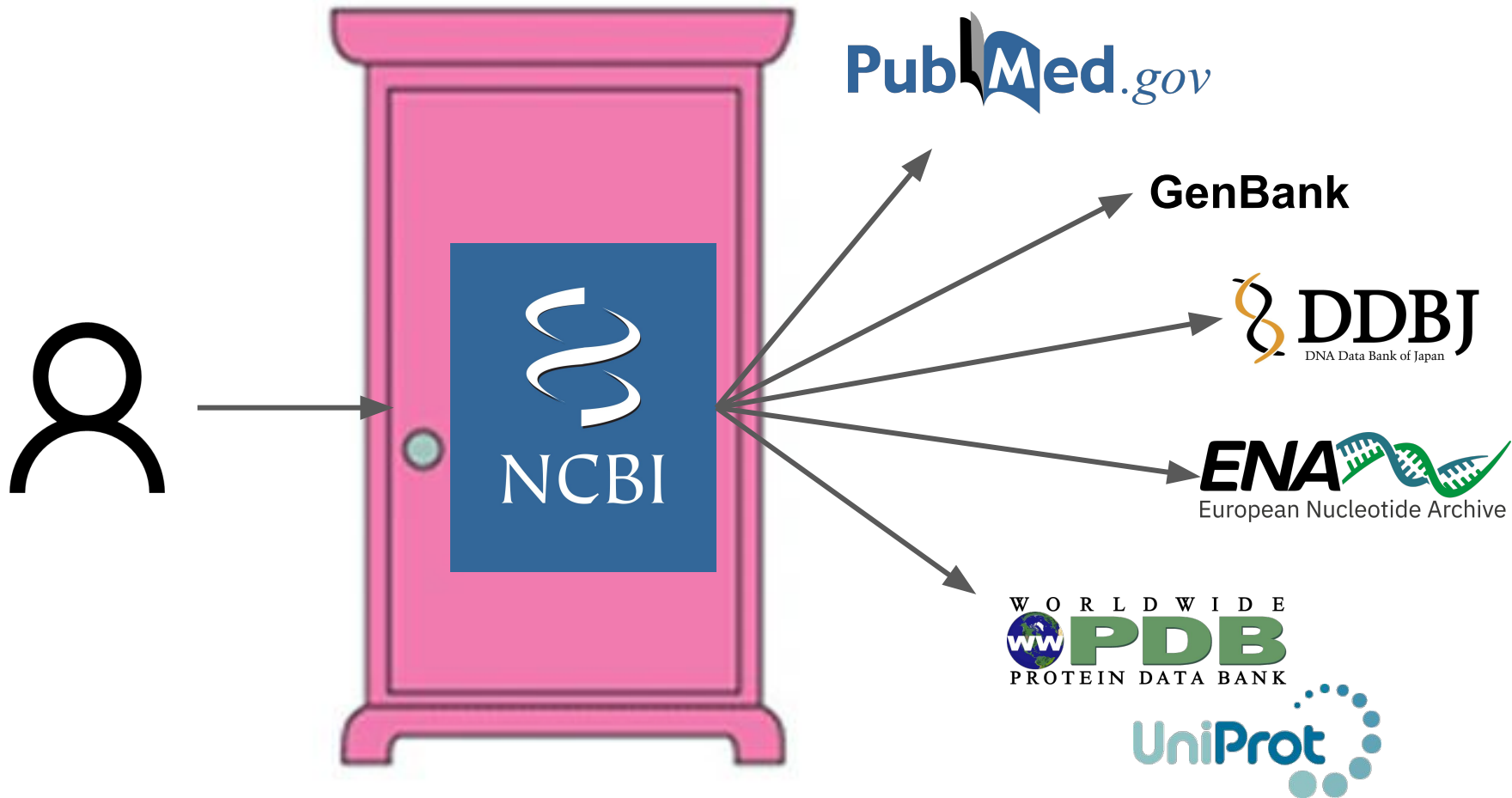
How to store biological sequences?

- In Notebook (Ancient)
- In Hard disk (Past)
- In Cloud/Database (Recently)



Introduction of Sequence Database

- **The National Center for Biotechnology Information (NCBI)**
 - Famous Biological Sequence Database
- Act like a portal to **access all biological database** including DNA, Protein, and etc.
 - Nucleotide sequence Database
 - Sanger Sequencing
 - The International Nucleotide Sequence Database Collaboration (INSDC)
 - GenBank (NCBI), EMBL-EBI, and DDBJ
 - NGS
 - Sequence Read Archive (SRA)
 - Protein sequence and structure
 - Protein Data Bank (PDB) and Uniport
 - PubMed database
 - Abstract and Citation of scientific articles
- Provide **web-based tools** for sequence searching and analysis



Web-based Tools for Bioinformatics

- BLAST (Basic Local Alignment Search Tool)
- COBALT
- ETC

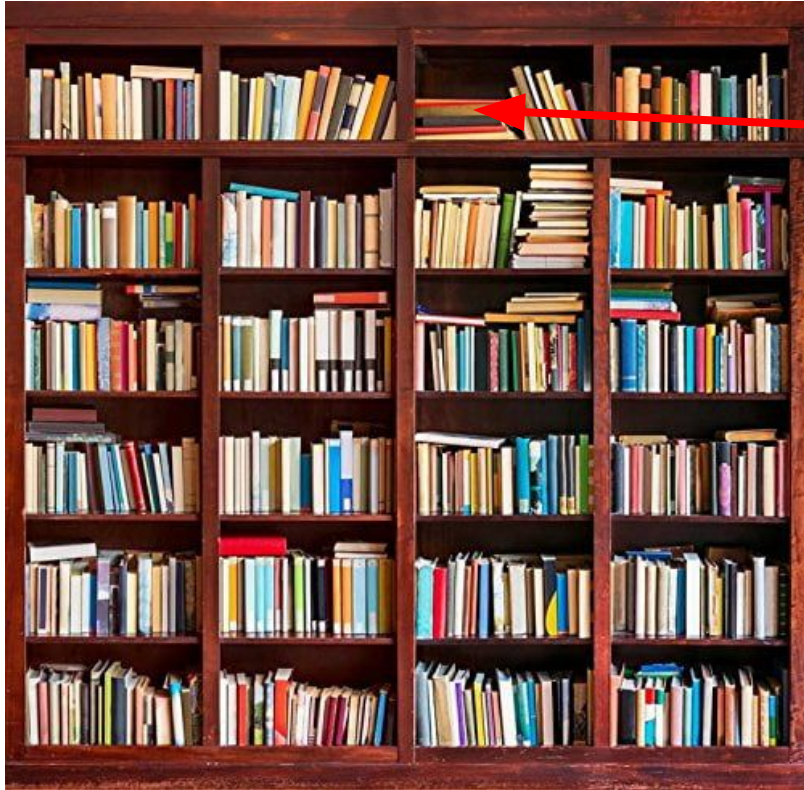
Analyze

NCBI provides a wide variety of data analysis tools that allow users to manipulate, align, visualize and evaluate biological data.

Selected Analysis Tools

All Tools	Literature	Health	Genomes	Genes	Proteins	Chemicals
Filter this table						
Tools		Description				
Amino Acid Explorer		Explores amino acid properties, substitutions and functions				
Assembly Archive		Links the raw sequence information found in the Trace Archive with assembly information found in GenBank/EMBL/DBJ				
Basic Local Alignment Search Tool (BLAST)		Finds regions of local similarity between biological sequences				
Batch Entrez		Retrieves records specified in an uploaded file of identifiers				
BioAssay Services		Tools that summarize the biological test results in the PubChem database				
BLAST Link (BLink)		Displays the results of a pre-computed BLAST search of a protein against all other protein sequences at NCBI				
BLAST Microbial Genomes		Finds regions of local similarity between query sequences and sequences from complete microbial genomes				
BLAST RefSeqGene		Finds regions of local similarity between query sequences and genomic sequences in the RefSeqGene/LRG set				
Comparative Genome Viewer (CGV)		Compare genomes based on whole genome assembly-assembly alignments				
Multiple Comparative Genome Viewer (MCGV)		MCGV is a graphical display for whole genome multiple sequence alignments.				
Cn3D		Displays and manipulates 3-dimensional structures and alignments from the Structure database				
COBALT		Performs protein multiple sequence alignments				
Concise Microbial Protein BLAST		Finds regions of local similarity between query proteins and proteins from complete microbial (prokaryotic) genomes				

How to search biological information and sequence?



I want this.



Search Systems in NCBI

- To Find the interesting things in NCBI-related database (Mostly Gene)
- Two Main Search Systems on NCBI
 - **Entrez** search systems (Like Google): keyword search
 - **Basic Local Alignment Search Tool (BLAST)**: sequence search

National Library of Medicine
National Center for Biotechnology Information

All Databases ▾

Welcome to NCBI

The National Center for Biotechnology Information advances science biomedical and genomic information.

Entrez

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

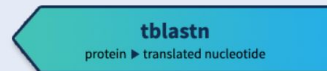
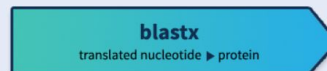
BLAST+ 2.13.0 is here!

Starting with this release, we are including the blastn_vdb and tblastn_vdb executables in the BLAST+ distribution.

Thu, 17 March 2022

[More BLAST news...](#)

Web BLAST

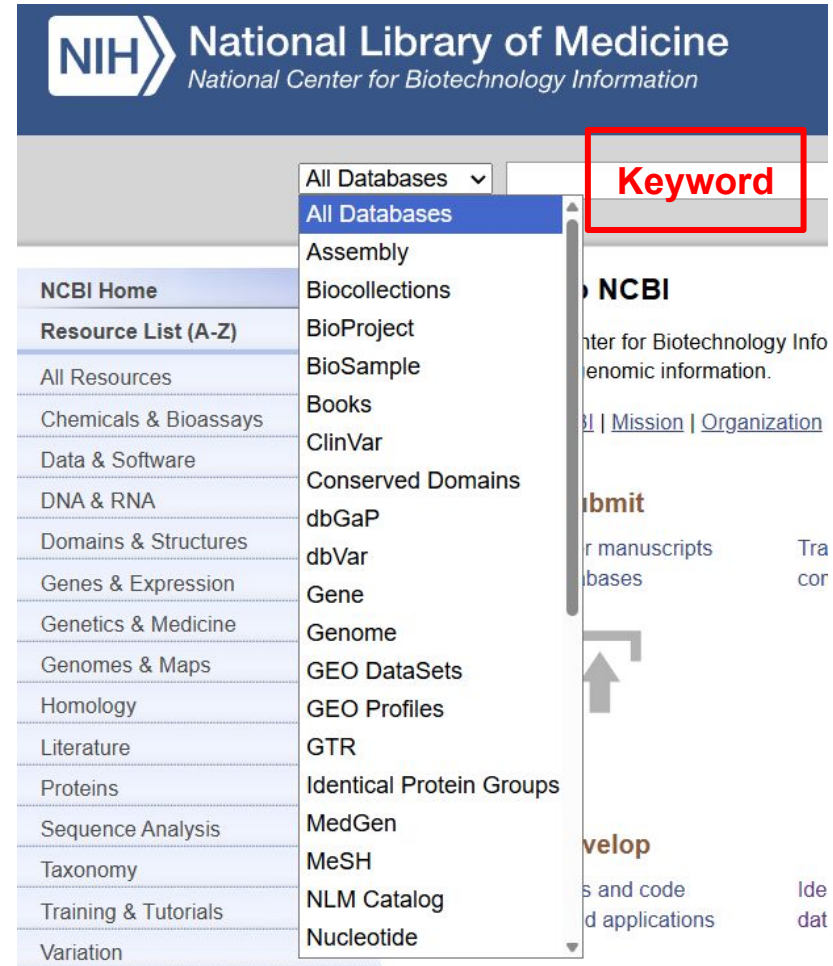


BLAST

Entrez Search System

- General Search to obtain Interesting Information from NCBI
- Narrow down your interesting information
- Usually start with the name of interesting gene, protein, or etc.

First Example: Search Well-Known Gene to obtain DNA, mRNA, and Protein sequence



Entrez Search Results in All Databases

Results found in 25 databases

Literature		
Bookshelf		904
MeSH		29
NLM Catalog		611
PubMed		174,948
PubMed Central		96,192
Genes		
Gene		9,628
GEO DataSets		11,149
GEO Profiles		2,500
Proteins		
Conserved Domains		69
Identical Protein Groups		97,774
Protein		444,139
Protein Family Models		88
Structure		862
Genomes		
Assembly / Genome	NCBI Datasets	
BioCollections		0
BioProject		558
BioSample		4,513
Nucleotide		87,215
SRA		82,113
Taxonomy		0
Clinical		
ClinicalTrials.gov		0
ClinVar		2,836
dbGaP		3
dbSNP		0
dbVar		495
GTR		69
MedGen		54
OMIM		173
PubChem		
BioAssays		302
Compounds		0
Pathways		0
Substances		76

BLAST Search System

- Basic Local Alignment Search Tool
 - BLAST Program
 - Special Program for sequence searching

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

BLAST+ 2.13.0 is here!

Starting with this release, we are including the blastn_vdb and tblastn_vdb executables in the BLAST+ distribution.

Thu, 17 March 2022 [More BLAST news...](#)

Web BLAST



Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide



Protein BLAST
protein ► protein

blastnblasttblasttblastx

Standard Nucleotide BLAST

BLASTn programs search nucleotide databases using a nucleotide query. more...

Enter Query Sequence

Enter accession number(s), g(s), or FASTA sequence(s) [Clear](#)

Query subrange [?](#)

TGGAACAGTGTCTACCAAGCATCTGCTCCCTCTACAGCTGGAGAACTAC
TGCAACTAGATGCAGCCCGCAGGCGAGCCCGCCACACCCCTCCGCTCTGCA
CCGAGAGACATGGAATAAAGCCCTGAACCCAGC

From
To

Or, upload file No file chosen [?](#)

Job Title
Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.) ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

Organism [?](#)
Optional ☐ exclude
Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to ☐ Sequences from type material

Entrez Query
Optional Enter an Entrez query to limit search [?](#) [YouTube](#) [Create custom database](#)

Program Selection

Optimize for ☒ Highly similar sequences (megablast)
☐ More dissimilar sequences (discontiguous megablast)
☐ Somewhat similar sequences (blastn)
Choose a BLAST algorithm [?](#)

BLAST

Search database core_nt using Megablast (Optimize for highly similar sequences)
☐ Show results in a new window

+ Algorithm parameters

BLAST Search System

- Search tool for **interesting sequence**
- Require **sequences** as the input to search

```
GAACTCGGGAAGCCGCGCAGAGAAGTGTGAGGCCGCGG
TAGGGCCGCATCCCGCTCCGGAGAGAAGTCTGAG
TCCGCCAGGCTCTGCAGGCCCGCGGAAGCTCGGTAA
TGATAAGCACGCCGGCCACTTTGCAGGGCGTCAC
CGCCTACACGCCCCCTCGTCTCTCGGACGGCGGCGT
CTAGCCTCGGGGCGCTCGGCCGCCCCGCCCTCTC
CGGGGGAGGAATCAAGAAGAGACTGCCCAATAGGGC
CGGCTTGACCCGCGAACAGGCGAGGGTTCCCGGG
GGAGTGGCGCGGCAGAAGGCCCCGCCAGGAGCCGA
GGGACAGCCCAGAGGAGGCGTGCCACGCTGCCG
GCGGAAGTGAGCCCTCCGCGAGCGCGAGGCCGCG
CGGGGCAGGCGGGGAAACCGGACAGTAGGGGCGG
GGCCGGGCGGCGATGGGGATGCGGGAGCACTACGC
GGAGCTGCACCCGTGCCCCCGGAATTGGGGATG
CAGAGCAGCGGCAGCGGGTATGGCAGGCAGCCGGCG
GGCCGGCCTCCAGCGCAGGTGCCCCGAGAGGCAGG
GGCTGGCCTGGGATGCGCGCGCACCTGCCCTCGCCC
CGCCCCGCCCCGACGAGGGGTGGTGGCCGAGGCC
CCGCCCCGCGACGCTCGCCTGAGGCGGGTCCGCTC
```

```
MGRRGSAPGNRGLRGCEGRRRRSADSVMAEQ
VALSRTQVCGILREELFQGDFAHQSDTHIFIIMG
ASGDLAKKKIYPTIWFLFRDGLLPENTFIVGYAR
SRLTVADIRKQSEPFFKATPEEKLKLEDFFARN
YVAGQYDDAASYQRLNSHMNALHLGSQANRLFYL
ALPPTVYEAVTKNIHESCMSQIGWNRIIVEKPF
RDLQSSDRLSNHISLRFREDQIYRIDHYLGKEMV
QNLMLVLRFANRIFGPIWNRDNIACVILTFKEPF
TEGRGYFDEFGIIRDVMQNHLLQMLCLVAMEKPA
STNSDDVRDEKVKVLCISEVQANNVVLGQYVGN
PDGEGEATKGYLDDPTVPRGSTTATFAAVVLYVE
NERWDGVPFILRCGKALNERKAEVRLQFHDVAGD
IFHQQCKRNELVIRVQPNVAVYTKMMTKKPGMFF
NPEESELDTYGNRYKNVKLPDAYERLILDVFCG
SQMHFVRSDELREAWRIFTPLLHQIELEKPKPIP
YIYSGRGPTEADELMKRVGFQYEGTYKWVNPBKL
```

```
>NC_000023.11:c154547569-154531390
Homo sapiens chromosome X,
GRCh38.p14 Primary Assembly
GAACTCGGGAAGCCGCGCAGAGAAGTGTGAGGCCGCGG
TAGGGCCGCATCCCGCTCCGGAGAGAAGTCTGAG
TCCGCCAGGCTCTGCAGGCCCGCGGAAGCTCGGTAA
TGATAAGCACGCCGGCCACTTTGCAGGGCGTCAC
CGCCTACACGCCCCCTCGTCTCTCGGACGGCGGCGT
CTAGCCTCGGGGCGCTCGGCCGCCCCGCCCTCTC
CGGGGGAGGAATCAAGAAGAGACTGCCCAATAGGGC
CGGCTTGACCCGCGAACAGGCGAGGGTTCCCGGG
GGAGTGGCGCGGCAGAAGGCCCCGCCAGGAGCCGA
GGGACAGCCCAGAGGAGGCGTGCCACGCTGCCG
GCGGAAGTGAGCCCTCCGCGAGCGCGAGGCCGCG
CGGGGCAGGCGGGGAAACCGGACAGTAGGGGCGG
```


Sequence Format in NCBI

- GenBank format
- Fasta format

```
LOCUS      AY211954                750 bp    DNA        linear    MAM 24-JUL-2016
DEFINITION Dendrolagus matschiei BRCA1 (BRCA1) gene, partial cds.
ACCESSION  AY211954
VERSION    AY211954.1
KEYWORDS   .
SOURCE     Dendrolagus matschiei (Huon tree kangaroo)
  ORGANISM Dendrolagus matschiei
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Metatheria; Diprotodontia; Macropodidae; Dendrolagus.
REFERENCE  1 (bases 1 to 750)
AUTHORS    Ramirez,C.J., Fleming,M.A., Potter,J.D., Ostrander,G.K. and
            Ostrander,E.A.
TITLE      Marsupial BRCA1: conserved regions in mammals and the potential
            effect of missense changes
JOURNAL     Oncogene 23 (9), 1780-1788 (2004)
PUBMED     15001988
```

```
>NC_000023.11:c154547569-154531390 Homo sapiens chromosome X, GRCh38.p14 Primary Assembly
GAACTCGGGAAGCCGCGCAGAAGTGTGAGGCCGCGGTAGGGCCGCATCCCGCTCCGGAGAGAAGTCTGAG
TCCGCCAGGCTCTGCAGGCCCGCGGAAGCTCGGTAATGATAAGCACGCCGGCCACTTTGCAGGGCGTCAC
CGCCTACACGCCCCCTCGTCTCTCGGACGGCGGCGTCTAGCCTCGGGGCGCTCGGCCGCCCCGCCCTCTC
CGGGGGAGGAATCAAGAAGAGACTGCCCAATAGGGCCGGCTTGACCCGCGAACAGGCGAGGGTTCCCGGG
GGAGTGGCGCGGCAGAAGGCCCCGCCAGGAGCCGAGGGACAGCCAGAGGAGGCGTGGCCACGCTGCCG
GCGGAAGTGGAGCCCTCCGCGAGCGCGGAGGCCGCCGGGGCAGGCGGGGAAACCGGACAGTAGGGGCGG
```

Fasta Format

- **Fasta:** Simple Formats for Sequences

Header ○ filename.**fasta**, filename.**fa**, filename.**fna**, and etc.

```
>NC_000023.11:c154547569-154531390 Homo sapiens chromosome X, GRCh38.p14 Primary Assembly
GAACTCGGGAAGCCGGCGAGAAGTGTGAGGCCGCGGTAGGGCCGCATCCCGCTCCGGAGAGAAGTCTGAG
TCCGCCAGGCTCTGCAGGCCCGCGGAAGCTCGGTAATGATAAGCACGCCGGCCACTTTGCAGGGCGTCAC
CGCCTACACGCCCCCTCGTCTCTCGGACGGCGGCGTCTAGCCTCGGGGCGCTCGGCCGCCCCGCCCTCTC
CGGGGGAGGAATCAAGAAGAGACTGCCCAATAGGGCCGGCTTGACCCGCGAACAGGCGAGGGTTCCCGGG
GGAGTGGCGCGGCAGAAGGCCCGCCAGGAGCCGAGGGACAGCCAGAGGAGGCGTGGCCACGCTGCCG
GCGGAAGTGGAGCCCTCCGCGAGCGCGCGAGGCCGCCGGGGCAGGCGGGGAAACCGGACAGTAGGGGCGG
```

Sequence

```
>Seq1 [organism=Carpodacus mexicanus] [clone=6b] actin (act) mRNA, partial cds
CCTTTATCTAATCTTTGGAGCATGAGCTGGCATAGTTGGAACCGCCCTCAGCCTCCTCATCCGTGCAGAATAATAATTTTCTTTATAGTAATACCAATCATGATCGGTGGT
TTCGGAAACTGACTAGTCCCCTCATAAT
>Seq2 [organism=uncultured bacillus sp.] [isolate=A2] corticotropin (CT) gene, complete cds
GGTAGGTACCGCCCTAAGNCTCCTAATCCGAGCAGAACTANGCCAACCCGGAGCCCTTCTGGGAGACGACTCAACACCACCTTCTTTGACCCAGCAGGAGGAGAGACCCA
GTACTATACCAGCACCTATTCTGATTCTT
>Seq3 [organism=Phalaenopsis equestris var. leucaspis]
CCTATACCTAATTTTCGGCGCATGAGCCGGAATGGTGGGTACCGCTCTAAGCCTCCTCATTCGAGCAGAACTAGGCCAACCCGGAGCCCTTCTGGGAGACGACCAAGTCTA
CAACGTGGTTGTACGGCCCATGCCTTCG
>Seq9 [organism=Petunia integrifolia subsp. inflata]
TAGTTGGAACAGCCCTCAGCCTACTCATCCGAGCAGAACTAGGCCAACCCGGAACCCCTCCTGGGAGATGACCAAATCTACAATGTAATCGTCACTGCCCATGCCTTCGTAA
TAATCTTCTTCATAGTAATACCAGTCATA
```

GenBank Format

- Locus
- Definition
- Accession
- Organism
- Reference
- Features
- Origin

LOCUS	AY211954	750 bp	DNA	linear	MAM 24-JUL-2016
DEFINITION	Dendrolagus matschiei BRCA1 (BRCA1) gene, partial cds.				
ACCESSION	AY211954				
VERSION	AY211954.1				
KEYWORDS	.				
SOURCE	Dendrolagus matschiei (Huon tree kangaroo)				
ORGANISM	<u>Dendrolagus matschiei</u> Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Metatheria; Diprotodontia; Macropodidae; Dendrolagus.				
REFERENCE	1 (bases 1 to 750)				
AUTHORS	Ramirez,C.J., Fleming,M.A., Potter,J.D., Ostrander,G.K. and Ostrander,E.A.				
TITLE	Marsupial BRCA1: conserved regions in mammals and the potential effect of missense changes				
JOURNAL	Oncogene 23 (9), 1780-1788 (2004)				
PUBMED	15001988				

FEATURES	Location/Qualifiers
source	1..750 /organism="Dendrolagus matschiei" /mol_type="genomic DNA" /db_xref="taxon:165202"
gene	<1..>750 /gene="BRCA1"
mRNA	<1..>750 /gene="BRCA1" /product="BRCA1"
CDS	<1..>750 /gene="BRCA1" /note="similar to Homo sapiens BRCA1 gene exon 11" /codon_start=1 /product="BRCA1" /protein_id=" AA031744.1 " /translation="VEPCGTDARATSLLPETTSLSPNTFQMNVEKAELCNKSKQPGLA KNQQSSLDSEKIECSAGKTLGAHELNAHHPCKERKELEDEPQHPSRGNPQNCQSGTK LKSSIQKVNEWLSRSDILISDNSYSRNHEQNAEMPSALEDGYPTADGNSNISEKTD PVADNNTYDPWPHVPERSCRPRAENNNIEDKIFGKTYRRKSGHPLNCITENLFAGAV VAPDSLIPPEPPKTTKLKHKRKTISNLQPDF"

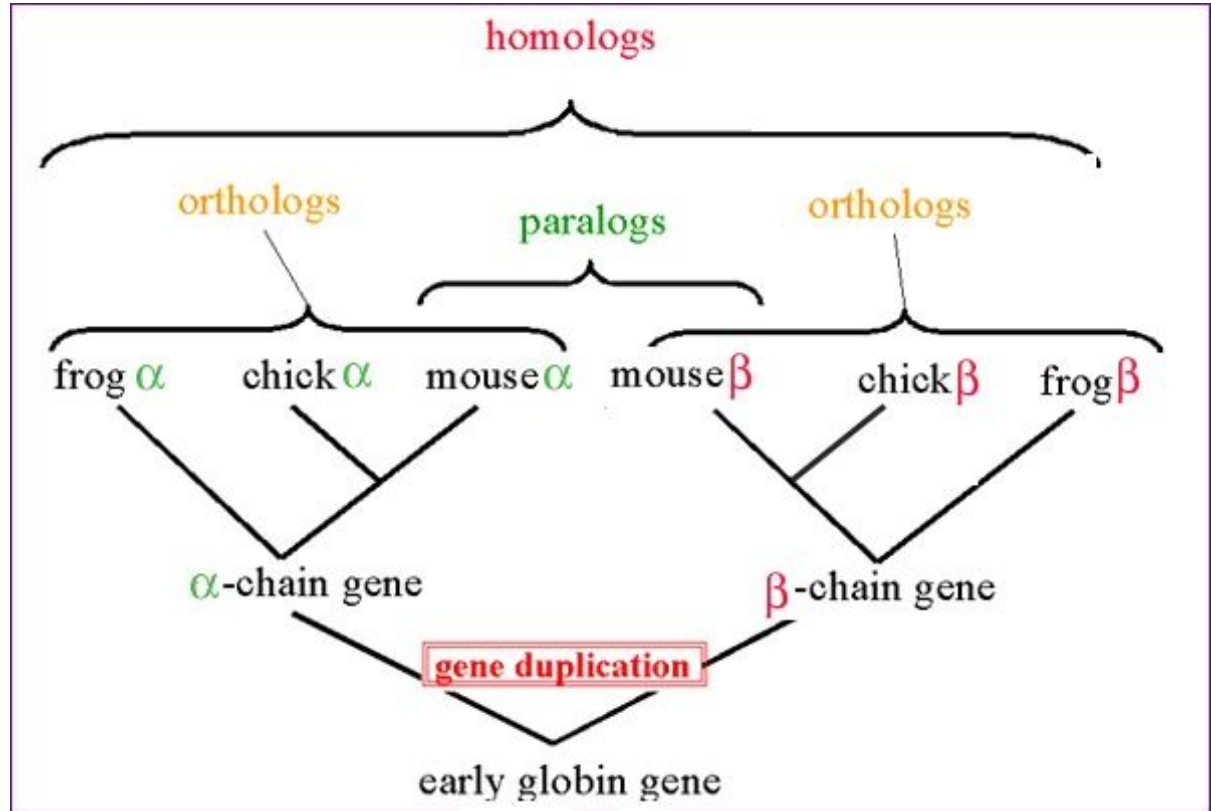
ORIGIN	1 gtggagccat gtggcacaga tgctcgtgcc acctcattac ttcctgaaac caccagctta 61 tcgccaaca cattccaaat gaatgtagaa aaggctgaac tctgtaataa aagcaaacag 121 cctggcttag caaaaaacca acagagcagt ctggatgaaa gtaaggaaat atgtagtgt 181 ggaaagaccc tgggtgccca tgagctgaat gccatcatc catgtgagag gaaagaacta 241 gaggatgagc cacagcacc tgagagccc agaggtaatc ctgagaactg ccagtctgga 301 accaaactga aaagtagtat tcagaaaagt aatgagtggg tatccaggag taatgatatt 361 ttaatttctg ataactccta tagtaggaac catgagcaga atgcagagat gcctagtgcc 421 ttagaagatg ggtatccaga tactgcagat ggaaattcta acatttctga gaagactgac 481 ccagtggctg acaacaacac ttatgatccc tggccacatg tgccctgaaag aagctgcccc 541 aggccagcag aaaaacaataa cattgaagat aaaatatttg gaaaaacctt tcggagaaaa 601 tcagggtcacc ctcatttgaa ttgtataact gaaaacttgt ttgctggagc tgttgttgt 661 cctgattctt tgatccctcc agagcctcca aaaaacaacca agttaaagca taaaagaaaa 721 accatatcta acctgcagcc tgaggatttc
//	

BLAST

Basic Local Alignment Search Tool

Background Knowledge: Homology

- **Homologs**
 - Homologous biological components
- **Orthologs**
 - Same function different species arose from ancestor
- **Paralogs**
 - Same species arose from duplication in ancestor

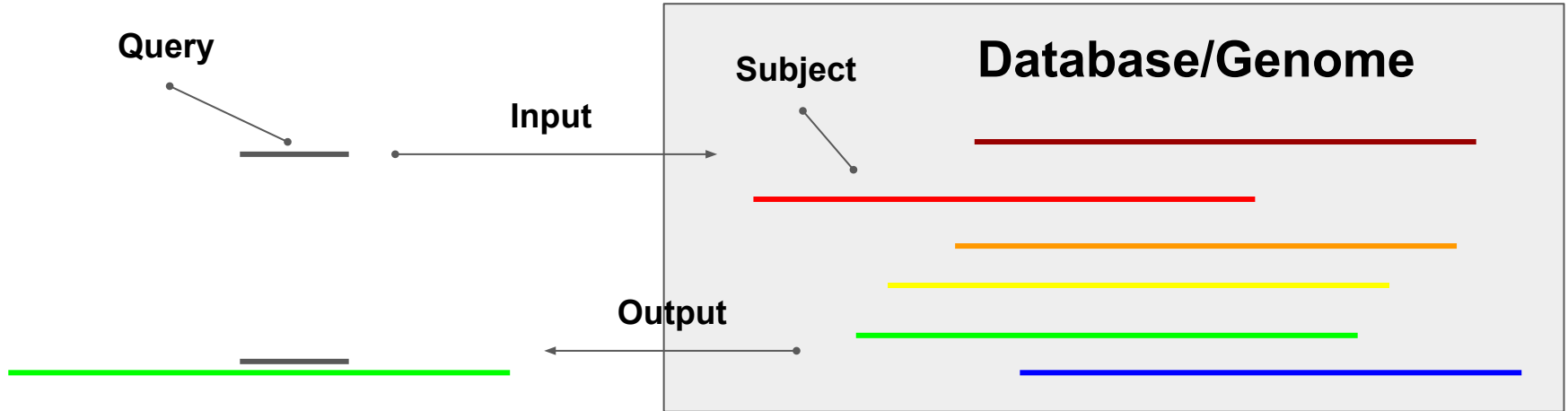


BLAST

- **Measure similarity between biological sequences**
 - **Non-chance similarities or Not**
- **If similarity do not due to chance then they must be due to something else**
 - **Identification** (Present or absent in database)
 - **Homology**: Evolutionary relationship -> related function
- **When you have a sequence, the possible questions are**
 - What is relate to?
 - **Function** of the sequences
 - **Homology: Orthologs** and **Paralogs**
 - Is it new or already in database?
 - Find the match sequences in database

BLAST Algorithm

- **Generate words from sequence**
- **Seed and Extent:** Search words and extent scoring in Database
- **Local Alignment:** Pairwise alignment between words and Database
- **Statistic Calculation:** Matching by **change** or **not**



BLAST Types

- **Basic Search**

- **blastn**
- **blastp**

- **Translated Search**

- **blastx**
- **tblastn**
- **tblastx**

Program	Query Type	Subject Type	Computation
blastn	N 	 N	~ 1X
blastp	P 	 P	~ 1X
blastx	N 	 P	~ 6X
tblastn	P 	 N	~ 6X
tblastx	N 	 N	~36X

(other BLAST types not listed: psiblast, deltablast, rpsblast)

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

NEWS

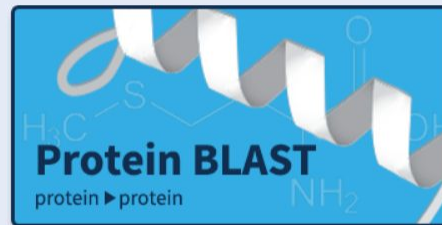
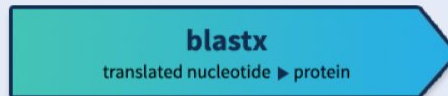
BLAST+ 2.13.0 is here!

Starting with this release, we are including the blastn_vdb and tblastn_vdb executables in the BLAST+ distribution.

Thu, 17 March 2022

 [More BLAST news...](#)

Web BLAST



Simple Searching of Nucleotide sequence

- blastn
 - Nucleotide Searching

Select Database

Specific organism

BLAST algorithm

The screenshot shows the NCBI BLAST search interface. At the top, there are tabs for different BLAST programs: **blastn**, blastp, blastx, tblastn, and tblastx. The **blastn** tab is selected. Below the tabs, the text "Standard Nucleotide BLAST" is visible. The main section is titled "Enter Query Sequence". It contains a large text input field labeled "Input sequence". To the right of this field are "Query subrange" options with "From" and "To" input boxes. Below the main input field, there are options to "Or, upload file" (with a "Choose File" button and "No file chosen" text) and a "Job Title" field. A checkbox "Align two or more sequences" is also present. The "Choose Search Set" section follows, with a "Database" dropdown menu set to "Core nucleotide database (core_nt)". Other options include "Organism" (with an "Add organism" button), "Exclude" (with checkboxes for "Models (XM/XP)" and "Uncultured/environmental sample sequences"), "Limit to" (with a checkbox for "Sequences from type material"), and "Entrez Query" (with a text input field and a "Create custom database" button). The "Program Selection" section at the bottom has an "Optimize for" dropdown menu set to "Highly similar sequences (megablast)". Other options include "More dissimilar sequences (discontiguous megablast)" and "Somewhat similar sequences (blastn)". A "Choose a BLAST algorithm" link is also present. At the very bottom, there is a "BLAST" button and a summary line: "Search database core_nt using Megablast (Optimize for highly similar sequences)". A checkbox "Show results in a new window" is also visible.

Standard Nucleotide BLAST

BLASTN programs search nucleotide databases using

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) Clear

Input sequence

Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database [?](#)

☒ Standard databases (nr etc.) ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

Core nucleotide database (core_nt) [?](#)

☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

☐ Sequences from type material

[YouTube](#) Create custom database

Enter an Entrez query to limit search [?](#)

Program Selection

Optimize for [?](#)

☒ Highly similar sequences (megablast)

☐ More dissimilar sequences (discontiguous megablast)

☐ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST

Search database core_nt using Megablast (Optimize for highly similar sequences)

☐ Show results in a new window

Nucleotide Database

- core_nt
- refseq_rna
- nr/nt

Choose Search Set

Database ☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript data

Organism Optional

Exclude Optional

Limit to Optional

Entrez Query Optional

Program Selection

Optimize for

BLAST

Algorithm parameters

General Parameters

Max target sequences

Short queries ☒ Automatically adjust parameters for short input sequences

Core nucleotide database (core_nt) ?

Core nucleotide database (core_nt)

RefSeq Select RNA sequences (refseq_select)

Reference RNA sequences (refseq_rna)

RefSeq Reference genomes (refseq_reference_genomes)

RefSeq Genome Database (refseq_genomes)

Nucleotide collection (nr/nt)

Whole-genome shotgun contigs (wgs)

Expressed sequence tags (est)

Sequence Read Archive (SRA)

Transcriptome Shotgun Assembly (TSA)

Targeted Loci(TLS)

High throughput genomic sequences (HTGS)

Patent sequences(pat)

PDB nucleotide database (pdb)

Human RefSeqGene sequences(RefSeq_Gene)

Genomic survey sequences (gss)

Sequence tagged sites (dbsts)

☐ exclude

be shown ?

ences

YouTube Cre

at similar sequences

BLAST Results

- **Max Score and Total score**
 - High score is better.
- **Query Cover: Percentage of query matching**
- **Expect (E-value): Lower is better**
 - This match is occurred by chance or others.
- **Per. Ident (Percentage identity)**
 - Percentage of Matched Sequence
- **Acc Len**
 - Accession length

Descriptions		Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments				Download	Select columns	Show	100		
☒ select all 100 sequences selected				GenBank	Graphics	Distance tree of results	MSA Viewer		
	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Glycine max superoxide dismutase [Cu-Zn] (LOC100305732), mRNA	Glycine max	1524	1524	100%	0.0	100.00%	825	NM_001249007.3
☒	PREDICTED: Glycine soja superoxide dismutase [Cu-Zn]-like (LOC114407659), mRNA	Glycine soja	1496	1496	98%	0.0	100.00%	897	XM_028370840.1
☒	Soybean clone JCVI-FLGm-2I5 unknown mRNA	Glycine max	1251	1251	83%	0.0	99.71%	685	BT089491.1
☒	PREDICTED: Glycine soja superoxide dismutase [Cu-Zn] (LOC114400541), mRNA	Glycine soja	1064	1064	91%	0.0	91.98%	879	XM_028363032.1
☒	Glycine max superoxide dismutase [Cu-Zn] (SOD1), mRNA	Glycine max	1007	1007	87%	0.0	91.94%	747	NM_001248369.2
☒	Glycine max cDNA, clone: GMFL01-18-E07	Glycine max	959	959	77%	0.0	93.90%	724	AK285900.1
☒	Soybean clone JCVI-FLGm-4D19 unknown mRNA	Glycine max	942	942	74%	0.0	94.45%	670	BT090454.1
☒	PREDICTED: Cajanus cajan superoxide dismutase [Cu-Zn] (LOC109794160), transcript variant X1, mR...	Cajanus cajan	852	852	88%	0.0	87.55%	865	XM_020353638.2
☒	PREDICTED: Cajanus cajan superoxide dismutase [Cu-Zn] (LOC109794160), transcript variant X2, mR...	Cajanus cajan	795	795	82%	0.0	87.61%	835	XM_020353639.2

BLAST Results

- **Score**
 - Higher is better
- **Expect (E-value): Lower is better**
 - Tell us about this match is occurred by chance or others
- **Identities**
 - Percentage of Matched Sequence
- **Gap**
 - Percentage of GAP

[Download](#) ▼

[GenBank](#) [Graphics](#)

Glycine max superoxide dismutase [Cu-Zn] (LOC100305732), mRNA

Sequence ID: [NM_001249007.3](#) Length: 825 Number of Matches: 1

Range 1: 1 to 825 [GenBank](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Identities	Gaps	Strand
1524 bits(825)	0.0	825/825(100%)	0/825(0%)	Plus/Plus
Query 1	CCATCCAtctttttcttcttactctagactcttccttttctcttctgtctAAGGGGTG	60		
Sbjct 1	CCATCCATCTTTTCTTCTTACTCTAGACTCTTCCTTTTCTTCTCTGTCTAAGGGGTG	60		
Query 61	CCCTGAGATCACATTGAACAATGGTGAAGGCTGTGGCCGTTCTTGGCAGCAGTGAGGGTG	120		
Sbjct 61	CCCTGAGATCACATTGAACAATGGTGAAGGCTGTGGCCGTTCTTGGCAGCAGTGAGGGTG	120		

Searching of amino acid sequence

- blastp program
 - Amino acid searching

Select Database

Specific organism

BLAST algorithm

BLAST® » blastp suite

blastn **blastp** blastx tblastn tblastx

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) Clear

Query subrange [?](#)

From

To

Input Sequence

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Databases ☒ Standard databases (nr etc.): ☐ Experimental databases

Compare ☐ Select to compare standard and experimental database [?](#)

Standard

Database [?](#)

Organism Optional ☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude Optional ☐ Models (XM/XP) ☐ Non-redundant RefSeq proteins (WP) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

[BLAST](#) Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

Protein Database

ClusteredNR, nr, or refseq_protein

The image shows a screenshot of the NCBI BLAST search interface. The 'Choose Search Set' dropdown menu is open, displaying a list of protein databases. The 'RefSeq Select proteins (refseq_select)' option is highlighted in blue. Other options include 'ClusteredNR (nr_cluster_seq)', 'Reference proteins (refseq_protein)', 'Model Organisms (landmark)', 'UniProtKB/Swiss-Prot (swissprot)', 'Non-redundant protein sequences (nr)', 'Patented protein sequences (pataa)', 'Protein Data Bank proteins (pdb)', 'Metagenomic proteins (env_nr)', and 'Transcriptome Shotgun Assembly proteins (tsa_nr)'. The 'BLAST' button is visible at the bottom left, and the '+ Algorithm parameters' link is at the bottom right.

Choose Search Set

Database

Organism
Optional

Program Selection

Algorithm

BLAST

+ Algorithm parameters

ClusteredNR (nr_cluster_seq) ▼

ClusteredNR (nr_cluster_seq)

RefSeq Select proteins (refseq_select)

Reference proteins (refseq_protein)

Model Organisms (landmark)

UniProtKB/Swiss-Prot (swissprot)

Non-redundant protein sequences (nr)

Patented protein sequences (pataa)

Protein Data Bank proteins (pdb)

Metagenomic proteins (env_nr)

Transcriptome Shotgun Assembly proteins (tsa_nr)

ed

Add organism

axa will be shown.

protein BLAST)

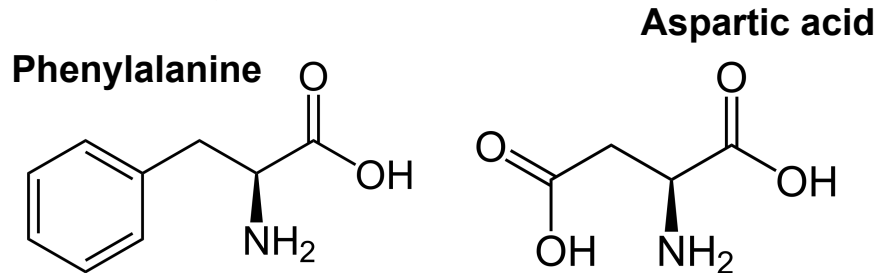
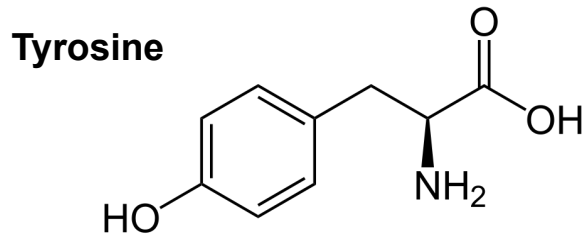
BLAST Results

- **Max Score and Total score**
 - High score is better.
- **Query Cover: Percentage of query matching**
- **Expect (E-value): Lower is better**
 - This match is occurred by chance or others.
- **Per. Ident (Percentage identity)**
 - Percentage of Matched Sequence
- **Acc Len**
 - Accession length

Clusters													
Graphic Summary			Alignments			Taxonomy							
Clusters producing significant alignments													
Download						Select columns			Show 100				
☒ select all 100 clusters selected													
GenPeptGraphicsDistance tree of resultsMultiple alignmentMSA Viewer													
	Cluster Composition		Cluster Ancestor		Cluster Representative Sequence		Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
	Click the  to see the cluster contents												
☒	28 member(s), 24 organism(s)		pea family		superoxide dismutase [Cu-Zn] [Glycine max]		285	285	100%	2e-96	94.74%	152	NP_001235298.1
☒	4 member(s), 4 organism(s)		eudicots		superoxide dismutase [Cu-Zn] [Senna tora]		276	276	100%	6e-93	90.13%	152	KAF7805166.1
☒	1 member(s), 1 organism(s)		eudicots		hypothetical protein Pint_20630 [Pistacia integerrima]		276	276	100%	6e-92	86.84%	228	KAJ0013852.1

BLAST Results

- **Score:** Higher is better
 - Bit score is a normalized score in search space.
- **Expect (E-value):** Lower is better
 - Tell us about this match is occurred by chance or others
- **Identity**
 - Percentage of Matched Sequence
- **Positive**
 - Indicate a **conservative substitution** or substitutions that are **often observed in related proteins**.
- **Gap**
 - Percentage of GAP



Score	Expect	Method	Identities	Positives	Gaps
281 bits(718)	6e-95	Compositional matrix adjust.	137/146(94%)	141/146(96%)	0/146(0%)
Query	1	VHLTPEEKNAVTTTLWGKVNVDVGGGALGRLLLVYPWTQRRFFESFGDLSSPDVAVMGPNKV			60
Sbjct	1	VHLTPEEK+AVT LWGKVNVDVGGGALGRLL+VYPWTQRRFFESFGDLS+PDVAVMGPNKV			60
Query	61	KAHGKKVLGAFSDGLNHLNLTGTFATLSELHCDKLHVDPENFKLLGNVLVLCVLAHHFGK			120
Sbjct	61	KAHGKKVLGAFSDGLAHLNLTGTFATLSELHCDKLHVDPENFRLGNVLVGLAHHFGK			120
Query	121	EFTPPVQAAYQKVVAGVANALAHKYH	146		
Sbjct	121	EFTPPVQAAYQKVVAGVANALAHKYH	146		

END